

The Invertible Matrix Theorem

Theorem (The many equivalences of being invertible)

If A is an $n \times n$ matrix then the following are equivalent:

- (a) *A is invertible.*
- (b) *A can be row reduced to the identity matrix.*
- (c) *A has n pivot positions.*
- (d) *$Ax = 0$ has only the trivial solution.*
- (e) *The columns of A form a linearly independent set.*
- (f) *The linear transformation $x \mapsto Ax$ is one-to-one.*
- (g) *The columns of A span $\mathbb{R}^n$.*
- (h) *The linear transformation $x \mapsto Ax$ is onto.*
- (i) *There is an $n \times n$ matrix C such that $CA = I$.*
- (j) *There is an $n \times n$ matrix D such that $AD = I$.*
- (k) *A^T is invertible.*

Rapid fire T/F. Fix an $n \times n$ matrix A .

- ① If A is invertible then the rows of A^{-1} are lin. indep.

True. ① A^{-1} is invertible since A is

② The rows of A^{-1} are the columns of $(A^{-1})^T$

③ $(A^{-1})^T$ is invertible so the columns are lin. indep.

② If the columns of A span $\mathbb{R}^n$ then $Ax = 0$ has a non-trivial solution.

False. A is invertible because its columns span $\mathbb{R}^n$ and so $Ax = 0$ has only trivial solutions.

③ if $Ax=b$ has a solution for all $b \in \mathbb{R}^n$
then A is invertible.

True. T_A is onto because $Ax=b$
has a solution for all b , and so
 A is invertible by the IMT.

Some consequences of the IMT.

Ex: ① Is
$$\begin{pmatrix} 1 & 4 & -3 & 7 \\ 0 & -1 & 2 & 5 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$
 invertible?

Yes! There is a pivot in each row.

② Is
$$\begin{pmatrix} 1 & -1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 4 \end{pmatrix}$$
 invertible?

No! 2nd column has no pivot.

Generalize this: We call a matrix upper triangular if all entries below the diagonal are 0.

Ex:

$$\begin{pmatrix} 1 & -1 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

upper
triangular.

$$\begin{pmatrix} 1 & -1 \\ \textcircled{1} & 2 \end{pmatrix}$$

← not triangular.

What upper triangular matrices are invert?

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \leftarrow \text{not invert.}$$

If the diagonal entries of an upper triangular matrix are all non-zero then the matrix is invertible.

We call a $n \times n$ ^{square} matrix lower triangular if all entries above the diagonal are 0.

Aside: Every square matrix is the product of a lower triangular matrix with an upper triangular matrix, (LU-decomposition).

How can we tell if a lower Δ matrix
is invertible?

All diagonal entries are non-zero.

The transpose is upper triangular.

We know that if A and B are $n \times n$ invertible matrices then AB is also invertible (with inverse $B^{-1}A^{-1}$).

Suppose we know AB is invertible.

Claim: Then both A and B are invertible.

Why? ① Show B is invertible.

What are solutions to $Bx = 0$?

If $Bx = 0$ then $(AB)x = 0$ and AB is invertible so by IMT $x = 0$.

Why is A invertible? I know AB is invertible so there is some C s.t.

$(AB)C = I$. So for every $b \in \mathbb{R}^n$

$A(\underline{BCb}) = b$ i.e. $Ax = b$ has a solution for every $b \in \mathbb{R}^n$.

So by the IMT, A is invertible.

One slide on partition matrices, section 2.4

$$A = \begin{array}{c} m_1 \\ m_2 \end{array} \left[\begin{array}{c|c} A_{11} & A_{12} \\ \hline A_{21} & A_{22} \end{array} \right] \begin{array}{c} n_1 \quad n_2 \end{array} \quad \text{and} \quad B = \begin{array}{c} k \\ n_1 \end{array} \left[\begin{array}{c} B_1 \\ B_2 \end{array} \right] \begin{array}{c} \leftarrow \\ \leftarrow \end{array}$$

$$A \text{ is } (m_1 + m_2) \times (n_1 + n_2)$$

$$AB = \left(\begin{array}{c} \underbrace{A_{11}B_1 + A_{12}B_2}_{\leftarrow m_1} \\ \underbrace{A_{21}B_1 + A_{22}B_2}_{\leftarrow m_2} \end{array} \right) \begin{array}{c} \leftarrow n_1 \\ \leftarrow n_2 \end{array}$$

$$(m_1 + m_2) \times k.$$

$$A = \begin{pmatrix} 2 \boxed{\frac{1}{2}} & 0 & 0 & 0 \\ 0 & 2 \boxed{\frac{1}{2}} & 0 & 0 \\ 0 & 0 & 2 \boxed{\frac{1}{2}} & 0 \\ 0 & 0 & 0 & 2 \boxed{\frac{1}{2}} \end{pmatrix} \leftarrow \text{block diagonal}$$

$Ax \rightarrow$ really ≈ 4

$$x \in \mathbb{R}^8$$

2×2 matrices mult.
by elements of $\mathbb{R}^2$.

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

where each $x_i \in \mathbb{R}^2$